An Introductory course of Particle Physics — Palashh B. Pal

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Contents

Chapter 1

Parity and Charge Conjugation

1.1 Exercise 6.1

(a)

We use notation $\mathbf{r} = (x, y, z)$. Now $\mathbf{r} \to \mathbf{r}' = (-x, -y, z)$. This we can write in terms of rotation as

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \cos\pi & \sin\pi & 0 \\ -\sin\pi & \cos\pi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Hence it is not discrete operation.

(b)

$\mathbf{r} \to \mathbf{r}' = (-x, y, z)$. This we can write as

$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\pi & \sin\pi \\ 0 & -\sin\pi & \cos\pi \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Which is Parity followed by rotation along x axis.

1.2 Exercise 6.2

(a)

Here $\mathbb{P} = \gamma_0\gamma_5$ and hence $\mathbb{P}^\dagger = \gamma_5\gamma_0$. Since the fermions are massless from eq. 6.21 we need to show that $\mathbb{P}$ satisfies

$$\mathbb{P}^\dagger\mathbb{P} = \mathbb{1}$$
$$\gamma_0\mathbb{P}^\dagger\gamma_0\vec{\gamma}\mathbb{P} = -\vec{\gamma}$$

First one is trivial and second can proved using $\{\gamma_5, \gamma_\mu\} = 0$

(b)

for $\mathcal{L}_{\text{int}} = \bar{\psi}\psi\phi$ we have using $\bar{\psi}_P(x) = \bar{\psi}(\tilde{x})\gamma_0\mathbb{P}^\dagger\gamma_0$

$$\begin{aligned}\mathcal{P}\mathcal{L}_{\text{int}}\mathcal{P}^{-1} &= \bar{\psi}(\tilde{x})\gamma_0(\gamma_0\gamma_5)^\dagger\gamma_0\gamma_0\gamma_5\psi(\tilde{x})\eta_p^\phi\phi(\tilde{x}) \\ &= -\eta_p^\phi\bar{\psi}\psi\phi\end{aligned}$$

hence $\eta_p^\phi = -1$ and the ϕ is pseudoscalar field.
for $\mathcal{L}_{\text{int}} = \bar{\psi}\gamma_5\psi\phi$

$$\begin{aligned}\mathcal{P}\mathcal{L}_{\text{int}}\mathcal{P}^{-1} &= \bar{\psi}(\tilde{x})\gamma_0(\gamma_0\gamma_5)^\dagger\gamma_0\gamma_5\gamma_0\gamma_5\psi(\tilde{x})\eta_p^\phi\phi(\tilde{x}) \\ &= \eta_p^\phi\bar{\psi}\psi\phi\end{aligned}$$

hence ϕ is intrinsic scalar

(c)

for $\mathcal{L}_{\text{int}} = h\bar{\psi}\psi\phi + h'\bar{\psi}\gamma_5\psi\phi$, the first part would require $\eta_p^\phi = -1$ while the second one $\eta_p^\phi = 1$ simultaneously and hence the interaction cannot be parity invariant.

1.3 Exercise 6.3

for the case of two different fields $F_P = \eta_2^*\eta_1\gamma_0F\gamma_0$. Hence the table would be same as that of (6.26) just with additional factor of $\eta_2^*\eta_1$

1.4 Exercise 6.4

1.5 Exercise 6.5

Lagrangian for scalar QED is

$$\begin{aligned}\mathcal{L}_{\text{int}} &= (D_\mu\phi)^\dagger(D^\mu\phi) - m^2\phi^\dagger\phi - \frac{1}{4}F^{\mu\nu}F_{\mu\nu} \\ &= (\partial_\mu\phi^* - ieA_\mu\phi^*)(\partial^\mu\phi + ieA^\mu\phi) - m^2\phi^\dagger\phi - \frac{1}{4}F^{\mu\nu}F_{\mu\nu} \\ &= (\partial_\mu\phi)^*(\partial^\mu\phi) + ie(\partial_\mu\phi)^*\phi A^\mu - ieA_\mu\phi^*(\partial^\mu\phi) + e^2A^2\phi^*\phi \\ &\quad - m^2\phi^\dagger\phi - \frac{1}{4}(F^{\mu\nu})^2\end{aligned}$$

We'll write

$$\begin{aligned}\mathcal{L}_{\text{int}} &= ie(\partial_\mu\phi)^*\phi A^\mu - ieA_\mu\phi^*(\partial^\mu\phi) + e^2A^2\phi^*\phi \\ &= ie[(\partial_0\phi)^*\phi A_0 - (\partial_i\phi)^*\phi A_i] - ie[A_0\phi^*(\partial_0\phi) - A_i\phi^*(\partial_i\phi)] \\ &\quad + e^2\phi^*\phi[A_0^2 - (A_i)^2]\end{aligned}$$

other terms are part of free theory and are parity invariant. Now

$$\begin{aligned}\mathcal{P}\mathcal{L}_{\text{int}}\mathcal{P}^{-1} &= ie[(\partial_0\phi(\tilde{x}))^*\phi(\tilde{x})(-\eta_p^\phi)A_0(\tilde{x}) + (\partial_i\phi(\tilde{x}))^*\phi(\tilde{x})\eta_p^\phi(\tilde{x})A_i(\tilde{x})] - \\ &\quad ie[(-\eta_p^\phi(\tilde{x}))A_0(\tilde{x})(\tilde{x})\phi^*(\tilde{x})(\partial_0\phi(\tilde{x})) + \eta_p^\phi A_i(\tilde{x})\phi^*(\tilde{x})(\partial_i\phi(\tilde{x}))] \\ &\quad + e^2\phi^*\phi[A_0(\tilde{x})^2 - (A_i(\tilde{x}))^2]\end{aligned}$$

Chapter 2

Non-Abelian Gauge Theories

2.1 Exercise 11.1

$$\begin{aligned}
\partial_\mu U^{-1} &= \partial_\mu (U^{-1} U U^{-1}) \\
&= \partial_\mu U^{-1} + U^{-1}(\partial_\mu U)U^{-1} + \partial_\mu U^{-1} \\
\implies \partial_\mu U^{-1} &= -U^{-1}(\partial_\mu U)U^{-1}
\end{aligned}$$

2.2 Exercise 11.2

$U = e^{-ieQ\theta}$

$$\begin{aligned}
A'_\mu &= A_\mu + \frac{i}{e}\left(\partial_\mu e^{-ieQ\theta}\right) e^{ieQ\theta} \\
&= A_\mu + Q\partial_\mu \theta
\end{aligned}$$

2.3 Exercise 11.3

$$\begin{aligned}
[D_\mu, D_\nu] f &= [\partial_\mu + ieA_\mu, \partial_\nu + ieA_\nu] f \\
&= (\partial_\mu + ieA_\mu)(\partial_\nu f + ieA_\nu f) - (\partial_\nu + ieA_\nu)(\partial_\mu f + ieA_\mu f) \\
&= ie(\partial_\mu A_\nu - \partial_\nu A_\mu) f \\
&= ieF_{\mu\nu} f
\end{aligned}$$

Chapter 3

Fermi Theory of Weak Interactions

3.1 Exercise 14.1

Lagrangian has mass dimension of 4. Fermions have mass dimension of 3/2. hence, $[G] = M^{-2}$

3.2 Exercise 14.2

$$\begin{aligned}\mathcal{L}_{\text{int}} &= -G\left[\bar{\psi}_p\gamma^\mu\psi_n\right]^\dagger\left[\bar{\psi}_e\gamma_\mu\psi_\nu\right] \\ &= -G\left[\bar{\psi}_n\gamma^\mu\psi_p\right]\left[\bar{\psi}_\nu\gamma_\mu\psi_e\right]\end{aligned} \tag{3.1}$$

Now we can see this Lagrangian terms creating n, ν and e^+ and annihilating p hence the process will be

$$p \to n + \nu + e^+ \tag{3.2}$$

which is β^+ decay

3.3 Exercise 14.3

The wavelength of the lepton will be

$$\lambda = \frac{hc}{pc} = \frac{hc}{E} \approx 10^{-13} \text{ m}$$

whereas the physical dimension of nucleus is 10^{-15} m. Hence the approximation is valid and good one

3.4 Exercise 14.4